

KINETICS & REACTOR DESIGN

CHE F311 Problems and Solutions I-Semester 2026-27

Every numerical question in the uploaded decks, with its solution.

Every question below is quoted from one of the uploaded files, and every solution uses the method taught in the deck that set it. The questions are grouped by source deck (Parts A–I). Where a slide prints its own answer the solution reproduces it; where a slide leaves something undefined, the Notes box says what was assumed. All numbers are re-computed by `krd-answers-check.py` kept beside this file.

Question inventory

ID	Source	Topic	Solved at
Q1	Ch1, slide 36	Rocket engine: rates of reaction from mole balance	p. ??
Q2	Ch1, slide 54	$A \rightarrow B$: CSTR, PFR and batch volumes/times, three rate laws	p. ??
Q3	Ch2, slides 19–24	Levenspiel sizing: CSTR and PFR at 80%	p. ??
Q4	Ch2, slides 31–36	Two CSTRs in series (40%, 80%)	p. ??
Q5	Ch2, slides 37–38	Two PFRs in series	p. ??
Q6	Ch2, slides 41–42	Mixed reactor, unknown stoichiometry	p. ??
Q7	Ch2, slides 45–46	PFRs in parallel branches: feed split	p. ??
Q8	Ch2, slide 59	P2-7 adiabatic $A + B \rightarrow C$ data, (a)–(e)	p. ??
Q9	Ch3, slides 5–7	Relative rates: $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$	p. ??
Q10	Ch3, slides 6–7	Relative rates: $2A + 3B \rightarrow 5C$	p. ??
Q11	Ch3, slides 42–44	Saponification stoichiometric table	p. ??
Q12	Ch3, slides 48–51	Flow-system stoichiometric table, $A + 2B \rightarrow C + D$	p. ??
Q13	Ch3 s.62 (+Ch5 s.77)	Dimerisation $2A \rightarrow R$ in MFR: rate equation	p. ??
Q14	Ch3, slide 63	Phosphine cracking: PFR size	p. ??
Q15	Ch4, slides 17–20	Ethylene-oxide batch data: k and batch time	p. ??
Q16	Ch4, slides 29–38	Ethylene glycol: CSTR volume, parallel and series 800-gal reactors	p. ??
Q17	Ch4, slides 44–45	Ethane cracking: PFR volume and number of tubes	p. ??
Q18	Ch4, slides 55–58	Gas $A + B \rightarrow 2C$ in PBR with pressure drop	p. ??
Q19	Ch4, slides 76–77	PBR + CSTR in series: order, exit pressure, particle size	p. ??
Q20	Ch4, slides 78–79	Series reactions $A \rightarrow B \rightarrow C$ in a batch reactor	p. ??
Q21	Ch4, slides 90–98	PFR with pressure drop: pipe length for 10%/20%	p. ??
Q22	Ch5, slides 12–17	Trityl + methanol: order and rate constant	p. ??
Q23	Ch5, slide 75	Dimethyl ether bomb: order and k	p. ??
Q24	Ch5, slide 72	First-order reversible $A \rightleftharpoons R$	p. ??
Q25	Ch5, slide 73	Half-order batch; $2A \rightarrow B$ volume change	p. ??
Q26	Ch5, slide 74	$A \rightarrow 3R$ in MFR: rate equation	p. ??
Q27	Ch5, slide 76	$A \rightarrow 2.5R$ batch, partial-pressure data	p. ??
Q28	Ch5, slides 78–79	Reaction bomb $2A \rightarrow B$ at 100 C	p. ??
Q29	Ch5, slide 80	Gambling tables (first-order analogue)	p. ??
Q30	Ch5, slide 81	$-r_A = kC_A^2$: $6 \times$ MFR and same-size PFR	p. ??
Q31	Ch5, slide 82	$A + 3B \rightarrow 6R$: X_A, X_B, C_B	p. ??
Q32	Ch5, slide 83	Gaseous feed with T, π change: $A + B \rightarrow 5R$	p. ??
Q33	Ch5, slide 84	Sucrose–sucrase Michaelis–Menten fit	p. ??
Q34	Ch5, slide 61	Reversible PFR: equilibrium and actual conversion	p. ??
Q35	Ch5, slide 62	$A \rightarrow R + 2S$ from total-pressure data	p. ??
Q36	Ch6, slide 32	Product distribution, 90% conversion, three contacting patterns	p. ??
Q37	Ch6, slide 33	Two MFRs in series, $r_R = k_1 C_A^2$, $r_S = k_2 C_A$	p. ??
Q38	Ch7, slides 14–17	n-Butane isomerisation: adiabatic PFR volume	p. ??
Q39	Ch7, slides 48–51	Adiabatic equilibrium + two interstage coolers	p. ??
Q40	Ch7, slides 57–59	Propylene-oxide CSTR operated adiabatically	p. ??
Q41	Ch7, slides 60–74	The same CSTR with a cooling coil	p. ??
Q42	Ch7, slide 95	Parallel reactions in a PFR with heat effects	p. ??
Q43	Catalysis, slide 48	Pellet concentration profile; diameter for $\eta = 0.8$	p. ??
Q44	Catalysis, slide 57	Internal diffusion: rates at 100/150 C, apparent E	p. ??
Q45	Non-ideal, s.17–19	Tracer pulse: $E(t)$ and fraction between 3 and 6 min	p. ??
Q46	Non-ideal, s.52–59	Dispersion model: conversion four ways	p. ??

The decks are photographs of slides; parts of some questions sit inside the pictures and are quoted as printed. Decks that repeat a question (the dimerisation of Q13 reappears in Ch5) are solved once and cross-referenced.

Part A – Mole balances and rates (Ch1-krd (1).pdf)

QQ1 – Rocket engine: rates of reaction. Ch1, slide 36

Source: Ch1-krd (1).pdf, slide 36 (the “Example” slide).

A rocket engine burns a stoichiometric mixture of fuel (liquid hydrogen) in oxidant (liquid oxygen). The combustion chamber is cylindrical, 75 cm long and 60 cm in diameter, and the combustion process produces 108 kg/s of exhaust gases. If combustion is complete, find the rate of reaction of hydrogen and of oxygen.

What “rate of reaction” means here. The rate is defined per unit extensive quantity of the system (Ch1 slides 30–34); for a homogeneous gas-phase combustion chamber the natural basis is the reactor *volume*, $-r_i = (\text{moles of } i \text{ reacting per unit time})/V$, with units $\text{mol}/(\text{m}^3 \cdot \text{s})$. So the whole problem is: find the molar rates at which H_2 and O_2 are consumed, and divide by the chamber volume.

Step 1 — the chamber volume. A cylinder of length $L = 0.75 \text{ m}$ and diameter $D = 0.60 \text{ m}$:

$$V = \frac{\pi}{4} D^2 L = \frac{\pi}{4} (0.60)^2 (0.75) = 0.2121 \text{ m}^3.$$

Step 2 — turn the exhaust mass flow into a molar flow. Complete combustion means every gram of exhaust leaves as water, $\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$, of molar mass 18 kg/kmol :

$$F_{\text{H}_2\text{O}} = \frac{108 \text{ kg/s}}{18 \text{ kg/kmol}} = 6.0 \text{ kmol/s}.$$

Step 3 — read the reactant rates off the stoichiometry. One mole of H_2 and half a mole of O_2 disappear per mole of water formed, so

$$F_{\text{H}_2, \text{reacted}} = 6.0 \text{ kmol/s}, \quad F_{\text{O}_2, \text{reacted}} = \frac{1}{2} (6.0) = 3.0 \text{ kmol/s}.$$

Step 4 — divide by the volume. At steady state with complete, uniform combustion the mole balance $\text{In} - \text{Out} + \text{Generation} = \text{Accumulation} (= 0)$ gives the consumption rate per unit volume directly:

$$-r_{\text{H}_2} = \frac{6.0}{0.2121} = 28.3 \text{ kmol m}^{-3} \text{ s}^{-1}, \quad -r_{\text{O}_2} = \frac{3.0}{0.2121} = 14.1 \text{ kmol m}^{-3} \text{ s}^{-1}.$$

Sanity checks for the exam: the two rates are in the same 2:1 ratio as the stoichiometric coefficients (this is the relative-rate rule of Part C), and the magnitudes sit in the “rocket engines” band of the rate spectrum printed on slide 35 — combustion is among the fastest reactions there are.

$$-r_{\text{H}_2} \approx 28.3 \text{ kmol}/(\text{m}^3 \cdot \text{s}) \approx 2.8 \times 10^4 \text{ mol}/(\text{m}^3 \cdot \text{s}); \quad -r_{\text{O}_2} \approx 14.1 \text{ kmol}/(\text{m}^3 \cdot \text{s}) \approx 1.4 \times 10^4 \text{ mol}/(\text{m}^3 \cdot \text{s}).$$

QQ2 – CSTR, PFR and batch for three rate laws. Ch1, slide 54

Source: Ch1-krd (1).pdf, slide 54 (the “Problem” slide).

The reaction $A \rightarrow B$ is to be carried out isothermally in a continuous-flow reactor. Calculate both the CSTR and the PFR volumes necessary to consume 99% of A ($C_A = 0.01 C_{A0}$) when the entering molar flow rate is $F_{A0} = 5 \text{ mol/h}$, assuming the rate laws (a) $-r_A = k$ with $k = 0.05 \text{ mol}/(\text{h} \cdot \text{dm}^3)$, (b) $-r_A = k C_A$ with $k = 0.0001 \text{ s}^{-1}$, (c) $-r_A = k C_A^2$ with $k = 3 \text{ dm}^3/(\text{mol} \cdot \text{h})$. The entering volumetric flow rate is $v_0 = 10 \text{ dm}^3/\text{h}$. (d) Repeat each case for a batch reactor reaching 99%.

Set-up used for every part. $X = 0.99$ and $C_{A0} = F_{A0}/v_0 = 5/10 = 0.5 \text{ mol/dm}^3$. The three design equations (slides 42–53):

$$\text{CSTR: } V = \frac{F_{A0} X}{(-r_A)_{\text{exit}}}, \quad \text{PFR: } V = F_{A0} \int_0^X \frac{dX}{-r_A}, \quad \text{batch: } t = N_{A0} \int_0^X \frac{dX}{-r_A V} = C_{A0} \int_0^X \frac{dX}{-r_A} \quad (V \text{ const}).$$

For a flow reactor with $-r_A = kC_A^n$ and $C_A = C_{A0}(1 - X)$, the integrals below are the standard ones; write them out in the exam rather than quoting results.

(a) Zero order, $-r_A = k = 0.05$. The rate is constant, so the CSTR exit rate equals the rate everywhere in the PFR and the two reactors coincide:

$$V_{\text{CSTR}} = \frac{F_{A0} X}{k} = \frac{5(0.99)}{0.05} = 99 \text{ dm}^3, \quad V_{\text{PFR}} = \frac{F_{A0}}{k} \int_0^{0.99} dX = 99 \text{ dm}^3.$$

$$\text{Batch: } t = \frac{C_{A0}}{k} \int_0^{0.99} dX = \frac{0.5(0.99)}{0.05} = 9.9 \text{ h}.$$

(b) First order, $-r_A = kC_A$. First fix the units: $k = 0.0001 \text{ s}^{-1} \times 3600 = 0.36 \text{ h}^{-1}$ — the single most common slip in this question. With $C_A = C_{A0}(1 - X)$:

$$\begin{aligned} V_{\text{CSTR}} &= \frac{F_{A0} X}{kC_{A0}(1 - X)} = \frac{v_0 X}{k(1 - X)} = \frac{10(0.99)}{0.36(0.01)} = 2750 \text{ dm}^3, \\ V_{\text{PFR}} &= \frac{F_{A0}}{kC_{A0}} \int_0^X \frac{dX}{1 - X} = \frac{v_0}{k} \ln \frac{1}{1 - X} = \frac{10}{0.36} (4.605) = 127.9 \text{ dm}^3, \\ t_{\text{batch}} &= \frac{1}{k} \ln \frac{1}{1 - X} = \frac{4.605}{0.36} = 12.8 \text{ h}. \end{aligned}$$

(c) Second order, $-r_A = kC_A^2 = kC_{A0}^2(1 - X)^2$.

$$\begin{aligned} V_{\text{CSTR}} &= \frac{F_{A0} X}{kC_{A0}^2(1 - X)^2} = \frac{v_0 X}{kC_{A0}(1 - X)^2} = \frac{10(0.99)}{3(0.5)(0.01)^2} = 66\,000 \text{ dm}^3, \\ V_{\text{PFR}} &= \frac{v_0}{kC_{A0}} \int_0^X \frac{dX}{(1 - X)^2} = \frac{v_0}{kC_{A0}} \cdot \frac{X}{1 - X} = \frac{10}{1.5} (99) = 660 \text{ dm}^3, \\ t_{\text{batch}} &= \frac{1}{kC_{A0}} \cdot \frac{X}{1 - X} = \frac{99}{1.5} = 66 \text{ h}. \end{aligned}$$

What the numbers teach (and what the deck's next section turns into the Levenspiel plot): as the order rises the CSTR volume explodes — $99 \rightarrow 2750 \rightarrow 66\,000 \text{ dm}^3$ — while the PFR grows mildly — $99 \rightarrow 128 \rightarrow 660 \text{ dm}^3$. The CSTR operates at the *exit* rate, i.e. at the smallest rate in the reactor, and punishing a small rate with a big volume is exactly what the design equations do. At 99% conversion a CSTR is $100\times$ the PFR for the second-order case.

(a) 99 dm^3 (CSTR = PFR), batch 9.9 h; (b) CSTR 2750, PFR 127.9 dm^3 , batch 12.8 h; (c) CSTR 66 000, PFR 660 dm^3 , batch 66 h.

Part B – Conversion and reactor sizing (Ch2-krd (1).pdf)

QQ3 – Sizing from the Levenspiel table. Ch2, slides 19–24

Source: Ch2-krd (1).pdf, slides 19–24 (the worked “CSTR/PFR sizing” problem).

Consider the isothermal gas-phase isomerization $A \rightarrow B$ ($T = 500 \text{ K}$, $P = 830 \text{ kPa} = 8.2 \text{ atm}$, charge pure A). The measured rates are:

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$-r_A \text{ (mol/m}^3\cdot\text{s)}$	0.45	0.37	0.30	0.195	0.113	0.079	0.05

Calculate the volumes required to achieve 80% conversion in a CSTR and in a PFR when species A enters at $F_{A0} = 0.4 \text{ mol/s}$.

Step 0 — build the Levenspiel column. The deck's method (slides 10–12) sizes reactors from the plot of $F_{A0}/(-r_A)$ vs X : the CSTR volume is a *rectangle* (height at exit conversion times width X); the PFR volume is the *area under the curve*. Dividing $F_{A0} = 0.4$ mol/s by each rate gives

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$F_{A0}/(-r_A)$ (m ³)	0.89	1.08	1.33	2.05	3.54	5.06	8.00

Step 1 — CSTR volume (slide 20). A CSTR operates at its *exit* conditions, so evaluate the rate at $X = 0.8$: $(1/-r_A)_{0.8} = 1/0.05 = 20$ m³·s/mol, i.e. $F_{A0}/(-r_A) = 8.00$ m³, and

$$V_{\text{CSTR}} = \left(\frac{F_{A0}}{-r_A} \right)_{X=0.8} X = 0.4 \frac{\text{mol}}{\text{s}} \times 20 \frac{\text{m}^3 \cdot \text{s}}{\text{mol}} \times 0.8 = 6.4 \text{ m}^3.$$

Step 2 — PFR volume by Simpson's rule (slides 21–23). $V_{\text{PFR}} = \int_0^{0.8} \frac{F_{A0}}{-r_A} dX$ has no closed form without a rate law, so the deck applies its five-point quadrature with $h = \Delta X = (0.8 - 0)/4 = 0.2$:

$$\begin{aligned} V &= \frac{\Delta X}{3} \left[\frac{F_{A0}}{-r_A}(0) + 4 \frac{F_{A0}}{-r_A}(0.2) + 2 \frac{F_{A0}}{-r_A}(0.4) + 4 \frac{F_{A0}}{-r_A}(0.6) + \frac{F_{A0}}{-r_A}(0.8) \right] \\ &= \frac{0.2}{3} [0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.00] = \frac{0.2}{3} (32.47) = 2.165 \text{ m}^3. \end{aligned}$$

Step 3 — read the picture (slide 24). The CSTR rectangle minus the PFR area is $6.4 - 2.165 = 4.235$ m³ of extra volume — the price of running the whole reactor at the slow exit rate.

Exam habit: write the two design equations, the $F_{A0}/(-r_A)$ table, the rectangle-vs-area argument, and the Simpson arithmetic. The deck's own printed answers are exactly 6.4 and 2.165 m³.

$$V_{\text{CSTR}} = 6.4 \text{ m}^3, V_{\text{PFR}} = 2.165 \text{ m}^3 \text{ (the PFR is } 4.235 \text{ m}^3 \text{ smaller).}$$

QQ4 — Two CSTRs in series. Ch2, slides 31–36

Source: Ch2-krd (1).pdf, slides 35–36.

For two CSTRs in series, 40% conversion is achieved in the first reactor. What is the volume of each reactor necessary to achieve 80% overall conversion of the entering species? (Same feed and rate data as Q3.)

The design equation for the n -th CSTR (slides 32–34). Balance A over reactor 1: In – Out + Generation = 0, i.e. $F_{A0} - F_{A1} + r_{A1}V_1 = 0$; with $F_{A1} = F_{A0} - F_{A0}X_1$ this gives $V_1 = F_{A0} \left(\frac{1}{-r_{A1}} \right) X_1$. Over reactor 2 the balance is $F_{A1} - F_{A2} + r_{A2}V_2 = 0$; substituting $F_{A1} = F_{A0}(1 - X_1)$ and $F_{A2} = F_{A0}(1 - X_2)$ leaves the difference of conversions:

$$V_2 = \frac{F_{A1} - F_{A2}}{-r_{A2}} = \frac{F_{A0}}{-r_{A2}} (X_2 - X_1).$$

Graphically each CSTR is a rectangle whose height is its *own exit* value of $F_{A0}/(-r_A)$.

Step 1 — reactor 1 ($X_1 = 0.4$, slide 36). $F_{A0}/(-r_A)|_{0.4} = 2.05$ m³:

$$V_1 = 2.05 \text{ m}^3 \times 0.4 = 0.82 \text{ m}^3.$$

Step 2 — reactor 2 ($X_1 = 0.4 \rightarrow X_2 = 0.8$). $F_{A0}/(-r_A)|_{0.8} = 8.00$ m³:

$$V_2 = 8.00 \text{ m}^3 \times (0.8 - 0.4) = 3.20 \text{ m}^3.$$

Step 3 — total and comparison. $V = V_1 + V_2 = 0.82 + 3.20 = 4.02$ m³, against 6.4 m³ for a single CSTR to 80%: splitting the conversion across two tanks saves 2.38 m³, because the first tank never has to operate at the slow $X = 0.8$ rate.

Exam habit: sketch the two Levenspiel rectangles (short one on $0 \leq X \leq 0.4$, tall one on $0.4 \leq X \leq 0.8$) and label the heights 2.05 and 8.00 m³ — the picture carries half the marks.

$$V_1 = 0.82 \text{ m}^3, V_2 = 3.20 \text{ m}^3, \text{ total } 4.02 \text{ m}^3 \text{ (vs } 6.4 \text{ m}^3 \text{ for one CSTR).}$$

QQ5 — Two PFRs in series. Ch2, slides 37–39

Source: Ch2-krd (1).pdf, slides 37–39.

Calculate the reactor volumes V_1 and V_2 for the plug-flow sequence shown when the intermediate conversion is 40% and the final conversion is 80% (same feed as Q3).

Why each PFR integrates over its own slice. Reactor 1 raises X from 0 to 0.4 and reactor 2 from 0.4 to 0.8; each integrates $F_{A0}/(-r_A)$ over that interval, so the two areas simply tile the area of one PFR (slide 44 proves that N PFRs in series give the same conversion as a single PFR of the total volume).

Step 1 — reactor 1 (slide 38), three-point Simpson, $h = 0.2$:

$$V_1 = \int_0^{0.4} \frac{F_{A0}}{-r_A} dX = \frac{0.2}{3} [0.89 + 4(1.33) + 2.05] = \frac{0.2}{3} (8.26) = 0.551 \text{ m}^3.$$

Step 2 — reactor 2 (slide 39), points 0.4, 0.6, 0.8:

$$V_2 = \int_{0.4}^{0.8} \frac{F_{A0}}{-r_A} dX = \frac{0.2}{3} [2.05 + 4(3.54) + 8.00] = \frac{0.2}{3} (24.21) = 1.614 \text{ m}^3.$$

Step 3 — total. $V = V_1 + V_2 = 0.551 + 1.614 = 2.165 \text{ m}^3$ — exactly the single-PFR volume of Q3, which is the point the deck is making: a PFR does not care where (or whether) you split it.

$V_1 = 0.551 \text{ m}^3$, $V_2 = 1.614 \text{ m}^3$, total $2.165 \text{ m}^3 =$ one PFR of Q3.

QQ6 – Mixed reactor, unknown stoichiometry. Ch2, slides 41–42

Source: Ch2-krd (1).pdf, slides 42–43 (the worked mixed-reactor problem).

One liter per minute of liquid containing A and B ($C_{A0} = 0.10$, $C_{B0} = 0.01$ mol/liter) flows into a mixed reactor of volume $V = 1$ liter. The materials react in a complex manner for which the stoichiometry is unknown. The outlet stream contains A, B and C with $C_A = 0.02$, $C_B = 0.03$, $C_C = 0.04$ mol/liter. Find the rate of reaction of A, B and C for the conditions within the reactor.

Why one balance per species. A CSTR at steady state balances each species on its own: In – Out + Generation = 0, $v_0 C_{i0} - v_0 C_i + r_i V = 0$, so with the space time $\tau = V/v_0 = 1/1 = 1$ min

$$r_i = \frac{C_i - C_{i0}}{\tau}.$$

Because the stream is liquid (constant volume), every species can be balanced independently — which is exactly why the question can be solved even though the stoichiometry is unknown.

Step 1 — evaluate the three balances.

$$r_A = \frac{0.02 - 0.10}{1} = -0.08, \quad r_B = \frac{0.03 - 0.01}{1} = +0.02, \quad r_C = \frac{0.04 - 0}{1} = +0.04 \text{ mol/(L} \cdot \text{min)}.$$

A is consumed; B and C are produced.

Step 2 — the ratios reveal the stoichiometry. Rates of reaction are proportional to the stoichiometric coefficients (the relative-rate rule of Part C): $-r_A : r_B : r_C = 0.08 : 0.02 : 0.04 = 4 : 1 : 2$, so the equation consistent with the measurements is $4A \rightarrow B + 2C$.

Exam phrasing: “the reactor is a CSTR, hence each species obeys $r_i = (C_i - C_{i0})/\tau$; the signs give consumption/production and the ratio of the rates gives the stoichiometric coefficients.”

$-r_A = 0.08$, $r_B = 0.02$, $r_C = 0.04$ mol/(L·min); the data imply $4A \rightarrow B + 2C$.

QQ7 – PFRs in two parallel branches. Ch2, slides 45–46

Source: Ch2-krd (1).pdf, slides 45–46.

The reactor setup consists of three plug-flow reactors in two parallel branches: branch D has a reactor of volume 50 liters followed by a reactor of volume 30 liters; branch E has a reactor of volume 40 liters. What fraction of the feed should go to branch D?

The rule (slide 45). PFRs in any parallel / parallel-series combination behave as a single PFR of the total volume *only if* the feed is distributed so that streams which meet have the same composition — i.e. the space time V/F must be the same for each parallel line; “any other way of feeding is less efficient”.

Step 1 — branch volumes. Series PFRs add (slide 44): branch D is $50 + 30 = 80$ L; branch E is 40 L.

Step 2 — equal space times. $V_D/F_D = V_E/F_E \Rightarrow F_D/F_E = 80/40 = 2$.

Step 3 — the fraction.

$$\frac{F_D}{F} = \frac{80}{80 + 40} = \frac{2}{3} \approx 0.67.$$

An earlier revision of these notes misread the figure as $20 + 30$ L and printed $5/9$; the slide prints 50 then 30 liters, so two-thirds is the value consistent with the deck.

$2/3 \approx 0.67$ of the feed to branch D (equal space time $80/80 = 40/40 = 1$ L per unit flow).

QQ8 – P2-7: adiabatic $A + B \rightarrow C$, parts (a)–(e). Ch2, slide 59

Source: Ch2-krd (1).pdf, slides 59–60 (“P2-7”).

The exothermic reaction $A \rightarrow B + C$ was carried out adiabatically and the following data were recorded; the entering molar flow rate of A was $F_{A0} = 300$ mol/min:

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A$ (mol/dm ³ ·min)	1.0	1.67	5.0	5.0	5.0	5.0	1.25	0.91

(a) PFR and CSTR volumes for 40%; (b) range of X where the two volumes are identical; (c) maximum X in a 105 dm³ CSTR; (d) X for a 72 dm³ PFR followed by a 24 dm³ CSTR; (e) X for the reverse order.

Step 0 — the Levenspiel column and its shape. $g(X) = F_{A0}/(-r_A) = 300, 179.6, 60, 60, 60, 60, 240, 329.7$ dm³ at the tabulated X . Plot it (slide 60): a V-shaped curve — adiabatic heating makes the rate *rise* at first ($1/(-r_A)$ falls to the flat bottom 60), then reactant depletion makes it climb steeply. All five parts are just rectangles (CSTR: $V = X g(X)$) and areas (PFR: $V = \int_0^X g dX$) on this curve.

(a) 40% conversion. CSTR rectangle at the exit height: $V = 0.4 \times 60 = 24$ dm³. PFR area by trapezia over 0, 0.2, 0.4: $V = 0.1(300 + 2 \times 179.6 + 60) = 71.9$ dm³ — here the PFR is *larger*, because the rate is slow at the entrance.

(b) Where CSTR = PFR. Track both with X : at $X = 0.6$, CSTR = $0.6 \times 60 = 36$ vs PFR = $71.9 + 0.2 \times 60 = 83.9$; at $X = 0.8$, CSTR = $0.8 \times 240 = 192$ vs PFR = $83.9 + \frac{0.2}{2}(60 + 240) = 113.9$. The rectangle overtakes the area between 0.6 and 0.8, at $X \approx 0.69$; below it the CSTR is the smaller reactor, above it the PFR is.

(c) 105 dm³ CSTR. Solve $X g(X) = 105$: $X = 0.6 \Rightarrow 36$, $X = 0.8 \Rightarrow 192$, interpolation on the rising limb gives $X \approx 0.70$.

(d) PFR then CSTR. The PFR area reaches 72 dm³ essentially at $X_1 = 0.40$ (part (a): 71.9). The CSTR then needs $(X_2 - X_1)g(X_2) = 24$: on the bottom/rising limb, $X_2 = 0.64$ gives $0.24 \times g(0.64) \approx 0.24 \times 96 \approx 23$ — so $X_2 \approx 0.64$.

(e) CSTR then PFR. $X_1 g(X_1) = 24$ has its useful root at $X_1 = 0.40$ (the height 60 rectangle). The PFR must add 72 dm³ from 0.40: the area $0.4 \rightarrow 0.6$ is 12, $0.6 \rightarrow 0.8$ is 15, $0.8 \rightarrow 0.9$ is 28.5 — 55.5 at the last datum, so the remaining area is eaten just beyond $X = 0.9$ where the data end; quote $X_2 \approx 0.90$.

The V-shaped curve is the exam point: at low X the CSTR wins (it sits on the flat bottom while the PFR drags through the slow entrance); past $X \approx 0.69$ the PFR wins. Order the units accordingly (slide 58: best arrangement rules).

(a) 24 dm³ / 71.9 dm³; (b) about $X = 0.69$; (c) $X \approx 0.70$; (d) $X \approx 0.64$; (e) $X \approx 0.90$.

Part C – Rate laws and stoichiometry (Ch3-krd.pdf)**QQ9 – Relative rates: NO oxidation.** Ch3, slides 5–7

Source: Ch3-krd.pdf, slides 5–7 (“Example”).

For $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$, NO_2 is formed at 4 mol/(m³·s). What are the rates of reaction of NO and of O₂?

The rule (slide 5). For $aA + bB \rightarrow cC + dD$ the stoichiometry ties every species rate together:

$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d},$$

i.e. divide each rate by its own coefficient (taken negative for reactants) and all four ratios equal the same single “rate of reaction”. Sign convention: $r_i > 0$ means formation, $r_i < 0$ consumption.

Step 1 — write the chain for this reaction ($a_{\text{NO}} = 2$, $b_{\text{O}_2} = 1$, $c_{\text{NO}_2} = 2$):

$$\frac{r_{\text{NO}}}{-2} = \frac{r_{\text{O}_2}}{-1} = \frac{r_{\text{NO}_2}}{+2}.$$

Step 2 — scale by the known rate. With $r_{\text{NO}_2} = +4$:

$$r_{\text{NO}} = \frac{-2}{+2}(4) = -4 \text{ mol}/(\text{m}^3 \cdot \text{s}), \quad r_{\text{O}_2} = \frac{-1}{+2}(4) = -2 \text{ mol}/(\text{m}^3 \cdot \text{s}).$$

Step 3 — check. $|r_{\text{NO}}| : |r_{\text{O}_2}| : |r_{\text{NO}_2}| = 4 : 2 : 4 = 2 : 1 : 2$ — the coefficients themselves, as the rule demands (this is the deck’s solution slide 7).

In the exam also state $-r_{\text{NO}} = 4$ mol/(m³·s): “NO is consumed as fast as NO₂ is formed; O₂ at half that rate.”

$r_{\text{NO}} = -4$, $r_{\text{O}_2} = -2$ mol/(m³·s).

QQ10 – Relative rates: $2A + 3B \rightarrow 5C$. Ch3, slides 6–7

Source: Ch3-krd.pdf, slides 6–7 (“Exercise”).

The reaction $2A + 3B \rightarrow 5C$ is carried out in a reactor. At a particular point the rate of disappearance of A is

10 mol/(dm³·s); what are the rates of B and C?

Step 1 — the chain. Same rule as Q9: $\frac{r_A}{-2} = \frac{r_B}{-3} = \frac{r_C}{5}$, and “disappearance of A is 10” means $r_A = -10$.
Step 2 — B (slide 7, left column).

$$r_B = \frac{-3}{-2} r_A = \frac{-3}{-2} (-10) = -15 \text{ mol}/(\text{dm}^3 \cdot \text{s}).$$

Step 3 — C (slide 7, right column).

$$r_C = \frac{5}{-2} r_A = \frac{5}{-2} (-10) = +25 \text{ mol}/(\text{dm}^3 \cdot \text{s}).$$

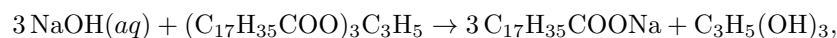
Check. 10 : 15 : 25 = 2 : 3 : 5 — the coefficients; B is consumed, C formed.

$$r_B = -15, r_C = +25 \text{ mol}/(\text{dm}^3 \cdot \text{s}).$$

QQ11 – Saponification stoichiometric table. Ch3, slides 42–44

Source: Ch3-krd.pdf, slides 41–44 (batch stoichiometric table + worked example).

Given the saponification for the formation of soap from aqueous caustic soda and glyceryl stearate,



i.e. $3A + B \rightarrow 3C + D$, let X be the conversion of sodium hydroxide. Set up a stoichiometric table expressing the concentration of *each species* in terms of its initial concentration and the conversion.

Step 1 — coefficients and changes (slide 41’s generic table). $a = 3$, $b = 1$, $c = 3$, $d = 1$. Per mole of A reacted the changes are $-N_{A0}X$ (A), $-\frac{b}{a}N_{A0}X = -\frac{1}{3}N_{A0}X$ (B), $+\frac{c}{a}N_{A0}X = +N_{A0}X$ (C), $+\frac{d}{a}N_{A0}X = +\frac{1}{3}N_{A0}X$ (D); with $\Theta_i = N_{i0}/N_{A0}$ the remaining moles are $N_i = N_{A0}(\Theta_i + \nu_i X/a)$.

Step 2 — liquid phase kills the volume term (slide 43). “We know that this is a liquid-phase reaction. Therefore $V = V_0$ ”, so $C_i = N_i/V_0$ and $N_{A0}/V_0 = C_{A0}$:

$$\begin{aligned} C_A &= C_{A0}(1 - X), & C_B &= C_{A0}(\Theta_B - \frac{1}{3}X), \\ C_C &= C_{A0}(\Theta_C + X), & C_D &= C_{A0}(\Theta_D + \frac{1}{3}X). \end{aligned}$$

This is exactly the deck’s filled table (slide 44).

Step 3 — what the feed gives. As fed there is no soap or glycerine, $\Theta_C = \Theta_D = 0$; if caustic soda and stearate are charged stoichiometrically $\Theta_B = \frac{1}{3}$, in which case B is exhausted exactly when $X = 1$.

Exam one-liner: “liquid phase $\Rightarrow V = V_0$, so divide the mole table by V_0 and replace N_{A0}/V_0 with C_{A0} .”

$$C_A = C_{A0}(1 - X), C_B = C_{A0}(\Theta_B - X/3), C_C = C_{A0}(\Theta_C + X), C_D = C_{A0}(\Theta_D + X/3).$$

QQ12 – Flow-system table for $A + 2B \rightarrow C + D$. Ch3, slides 48–52

Source: Ch3-krd.pdf, slides 49–52 (worked flow-table example).

Liquid phase, $A + 2B \rightarrow C + D$, with $C_{A0} = 1.8$ and $C_{B0} = 6.6 \text{ kmol}/\text{m}^3$ ($C_{C0} = C_{D0} = 0$). Take A as the basis and express every concentration in terms of X .

Step 1 — feed ratios and δ (slide 49). $\Theta_i = F_{i0}/F_{A0} = C_{i0}/C_{A0}$:

$$\Theta_B = \frac{6.6}{1.8} = 3.67, \quad \Theta_C = \Theta_D = 0,$$

and $\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 = 1 + 1 - 2 - 1 = -1$ (moles shrink), though for a *liquid* feed δ never changes the volume.

Step 2 — the molar-flow table (slide 50), changes $-F_{A0}X$, $-2F_{A0}X$, $+F_{A0}X$, $+F_{A0}X$:

Species	Effluent molar flow
A	$F_A = F_{A0}(1 - X)$
B	$F_B = F_{A0}(\Theta_B - 2X) = F_{A0}(3.67 - 2X)$
C	$F_C = F_{A0}X$
D	$F_D = F_{A0}X$

Step 3 — concentrations (slide 52). Liquid phase: $v = v_0$, so $C_i = F_i/v_0 = C_{A0}(\dots)$:

$$C_A = C_{A0}(1 - X), \quad C_B = C_{A0}(3.67 - 2X), \quad C_C = C_D = C_{A0}X.$$

The point of the exercise: even with $\delta = -1$ nothing expands, because liquids do not. The same reaction in the gas phase would need the $(1 + \varepsilon X)$ factor — contrast with Q14.

$$C_A = C_{A0}(1 - X), C_B = C_{A0}(3.67 - 2X), C_C = C_D = C_{A0}X.$$

QQ13 – Dimerisation $2A \rightarrow R$ in a MFR. Ch3, slide 62

Source: Ch3-krd.pdf, slide 62; repeated as Ch5-krd.pdf slide 77 (“L 5.2”).

Pure gaseous reactant A ($C_{A0} = 100$ millimol/liter) is fed at a steady rate into a mixed flow reactor ($V = 0.1$ liter) where it dimerises ($2A \rightarrow R$). Four runs:

Run	1	2	3	4
v_0 (liter/h)	10.0	3.0	1.2	0.5
C_A (mmol/L)	85.7	66.7	50	33.4

Find a rate equation for this reaction.

Step 1 — conversion per run (gas phase!). $2A \rightarrow R$ gives $\varepsilon_A = \frac{1-2}{2} = -0.5$, so $C_A = C_{A0} \frac{1-X}{1-0.5X}$; inverting, $X = \frac{C_{A0}-C_A}{C_{A0}-0.5C_A}$:

$$X = 0.250, 0.500, 0.667, 0.800.$$

(A constant-density formula would give different, wrong X here.)

Step 2 — the MFR balance gives one rate per run. $-r_A = \frac{F_{A0}X}{V} = \frac{v_0 C_{A0}X}{V}$; e.g. run 1: $\frac{10 \times 100 \times 0.25}{0.1} = 2500$ mmol/(L·h). All four:

$$-r_A = 2502, 1499, 800, 400 \text{ mmol/(L} \cdot \text{h)}.$$

Step 3 — find the order from the four $(C_A, -r_A)$ pairs. A log-log plot has slope $1.96 \approx 2$: second order in A. Check by division: $-r_A/C_A^2 = 0.34, 0.34, 0.32, 0.36$ — constant, so

$$-r_A \approx 0.34 C_A^2 \quad (C_A \text{ in mmol/L, } -r_A \text{ in mmol/(L} \cdot \text{h)}).$$

Method sentence for the exam: “the mixed reactor evaluates the rate at the exit conditions; four space velocities give four rates; the order is the log-log slope.”

Second order in A: $-r_A \approx 0.34 C_A^2$ (mmol, L, h).

QQ14 – Phosphine cracking PFR. Ch3, slide 63

Source: Ch3-krd.pdf, slide 63 (“Example”).

$4\text{PH}_3(g) \rightarrow \text{P}_4(g) + 6\text{H}_2(g)$ proceeds at 649°C with the first-order rate $-r_{\text{PH}_3} = \left(\frac{10}{\text{hr}}\right) C_{\text{PH}_3}$. What size of plug-flow reactor operating at 649°C and 460 kPa can produce 80% conversion of a feed of 40 mol/h of pure phosphine?

Step 1 — enter in concentrations. Ideal gas at the reactor state:

$$C_{A0} = \frac{P_0}{RT} = \frac{460\,000}{8.314 \times 922.15} = 60.0 \text{ mol/m}^3.$$

Step 2 — the expansion factor. $4 \rightarrow 10$ moles: $\varepsilon_A = \frac{10-4}{4} = 1.5$, so $C_A = C_{A0} \frac{1-X}{1+1.5X}$ — the gas *expands* as it cracks and the concentration falls faster than $(1-X)$.

Step 3 — PFR design equation, first order.

$$V = F_{A0} \int_0^{0.8} \frac{dX}{k C_A} \cdot \frac{1+1.5X}{1-X} = \frac{F_{A0}}{k C_{A0}} \left[(1+\varepsilon) \ln \frac{1}{1-X} - \varepsilon X \right].$$

Step 4 — substitute. $\frac{F_{A0}}{k C_{A0}} = \frac{40}{10 \times 60} = 0.0667 \text{ m}^3$; the bracket at $X = 0.8$: $2.5 \ln 5 - 1.5(0.8) = 4.024 - 1.2 = 2.824$:

$$V = 0.0667 \times 2.824 = 0.188 \text{ m}^3 (\approx 188 \text{ L}).$$

Two classic slips the deck warns about implicitly: forgetting ε (which would give $0.0667 \times \ln 5 = 0.107 \text{ m}^3$, far too small) and using C_{A0} at STP instead of at $649^\circ\text{C}/460 \text{ kPa}$.

$V \approx 0.19 \text{ m}^3$ (about 190 L).

Part D – Isothermal design (Ch4-Krd.pdf)

QQ15 – Ethylene-oxide batch data. Ch4, slides 17–20

Source: Ch4-Krd.pdf, slides 17–20.

500 mL of 2 M ethylene oxide mixed with 500 mL of water (0.9 wt% H_2SO_4 catalyst) at 55°C ; after mixing $C_{A0} = 1.0 \text{ kmol/m}^3$. Glycol concentration is recorded against time. Determine k , and the batch time for 90% conversion.

Step 1 — choose the order to test. Hydrolysis with water in vast excess is pseudo-first-order, so plot $\ln(C_{A0}/C_A)$ vs t .

Step 2 — the straight line. The printed $\ln$ column against t is linear; the deck's slope is $k = 0.311 \text{ min}^{-1}$ (a least-squares fit of the same printed points gives 0.315 — agree to quote).

Step 3 — batch time for 90%. First order:

$$t_{0.9} = \frac{1}{k} \ln \frac{1}{1 - 0.9} = \frac{2.303}{0.311} = 7.4 \text{ min.}$$

Step 4 — the deck's scaling slide. At a much smaller $k = 10^{-4} \text{ s}^{-1}$ the same formula gives $t_{0.9} = 2.303/10^{-4} = 23\,000 \text{ s} = 6.4 \text{ h}$ — the point being that the batch time scales exactly as $1/k$.

$$k \approx 0.311 \text{ min}^{-1}; t_{90} \approx 7.4 \text{ min (and 6.4 h if } k = 10^{-4} \text{ s}^{-1}\text{)}.$$

QQ16 – Ethylene glycol CSTR. Ch4, slides 29–38

Source: Ch4_Krd.pdf, slides 29–38.

Produce 200 million lb/yr of ethylene glycol at 80% conversion in a CSTR. A 1 lbmol/ft³ solution of ethylene oxide is mixed with an equal volume of water (0.9 wt% H₂SO₄); $k = 0.311 \text{ min}^{-1}$ (Q15). (a) Find the CSTR volume; (b) the conversion of two 800-gal reactors in parallel, each fed half; (c) in series.

Step 1 — molar flows. $F_C = \frac{200 \times 10^6}{62} = 3.226 \times 10^6 \text{ lbmol/yr} = 6.14 \text{ lbmol/min}$; at 80%, $F_{A0} = F_C/0.8 = 7.67 \text{ lbmol/min}$.

Step 2 — volumetric flows and C_{A0} . The oxide stream is 1 lbmol/ft³, so $v_{A0} = 7.67 \text{ ft}^3/\text{min}$; mixing with an equal water volume doubles it, $v_0 = 15.34 \text{ ft}^3/\text{min}$, halving the concentration, $C_{A0} = 0.5 \text{ lbmol/ft}^3$.

(a) **The CSTR volume.**

$$V = \frac{F_{A0}X}{kC_{A0}(1-X)} = \frac{7.67 \times 0.8}{0.311 \times 0.5 \times 0.2} = 197.3 \text{ ft}^3 \quad (\text{deck value}).$$

(b) **Two 800-gal reactors in parallel.** Each holds $800/7.48 = 107 \text{ ft}^3$ and sees only its half-stream $v_{A0} = 7.67 \text{ ft}^3/\text{min}$: $\tau = 107/7.67 = 13.94 \text{ min}$, Damköhler $Da = k\tau = 4.34$,

$$X = \frac{Da}{1 + Da} = 0.81.$$

(c) **The same two in series.** Now each sees the full $v_0 = 15.34$: $\tau = 6.97 \text{ min}$, $Da = 2.17$, and two CSTRs in series multiply:

$$X = 1 - \frac{1}{(1 + Da)^2} = 0.90 \quad (\text{deck values}).$$

Exam point of (b)/(c): splitting the feed between parallel CSTRs keeps each at the high feed concentration (good per unit volume); series staging keeps the first reactor off the slow exit rate.

$$(a) 197.3 \text{ ft}^3; (b) X \approx 0.81; (c) X \approx 0.90.$$

QQ17 – Ethane cracking PFR and tube count. Ch4, slides 44–45

Source: Ch4_Krd.pdf, slides 44–45 (MATLAB solution on slides 84–95).

Produce 300 million lb/yr of ethylene by cracking pure ethane ($\text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_4 + \text{H}_2$, irreversible elementary) at 80% conversion, 1100 K, 6 atm; $k = 0.072 \text{ s}^{-1}$ at 1000 K with $E = 82 \text{ kcal/mol}$. How many 40-ft PFR tubes of 0.0205 ft^3 in parallel?

Step 1 — k at the reactor temperature (Arrhenius).

$$k(1100) = 0.072 \exp \left[\frac{82\,000}{1.987} \left(\frac{1}{1000} - \frac{1}{1100} \right) \right] = 3.07 \text{ s}^{-1}.$$

Step 2 — enter in concentration and flow. $C_{A0} = \frac{P}{RT} = \frac{6 \times 101\,325}{8.314 \times 1100} = 66.5 \text{ mol/m}^3$; $F_{A0} = \frac{300 \times 10^6 \text{ lb/yr}}{28} = 154 \text{ mol/s}$; hence $v_0 = F_{A0}/C_{A0} = 2.32 \text{ m}^3/\text{s}$.

Step 3 — the expanding-gas PFR integral. $\varepsilon = 1$; first order:

$$\begin{aligned} V &= \frac{v_0}{k} \left[(1 + \varepsilon) \ln \frac{1}{1 - X} - \varepsilon X \right] = \frac{2.32}{3.07} [2 \ln 5 - 0.8] \\ &= 0.756 \times 2.419 = 1.83 \text{ m}^3 \quad (= 64.6 \text{ ft}^3). \end{aligned}$$

Step 4 — tube count. $64.6/0.0205 \approx 3.2 \times 10^3$ tubes in parallel.

The deck solves the same problem numerically in MATLAB (slides 84–95) without printing the final volume; the closed-form first-order gas-phase integral above is the method the deck derives on slides 42–43.

$V \approx 1.8 \text{ m}^3$ (65 ft³); about 3150 tubes of 0.0205 ft³ in parallel.

QQ18 – PBR with pressure drop, $A + B \rightarrow 2C$. Ch4, slides 55–58

Source: Ch4_Krd.pdf, slides 55–58.

Equimolar feed of $A + B \rightarrow 2C$ over a packed bed; $k = 1.5 \text{ dm}^6/(\text{mol} \cdot \text{dotkg})$, $\alpha = 0.0099 \text{ kg}^{-1}$, $\varepsilon = 0$. Find the conversion at $W = 100 \text{ kg}$ of catalyst.

Step 1 — rate law with the pressure factor. $\varepsilon = 0$ and an equimolar feed give $C_A = C_B = C_{A0}(1 - X)p$ with $p = (1 - \alpha W)^{1/2}$, so $-r_A = kC_A C_B = kC_{A0}^2(1 - X)^2(1 - \alpha W)$.

Step 2 — separate and integrate. The mole balance $\frac{dX}{dW} = \frac{kC_{A0}^2}{F_{A0}}(1 - X)^2(1 - \alpha W)$ separates cleanly:

$$\frac{X}{1 - X} = \frac{kC_{A0}^2}{F_{A0}} \left(W - \frac{\alpha W^2}{2} \right).$$

Step 3 — calibrate the unknown group from the deck. The deck prints $X = 0.75$ for the *no-drop* case, where the bracket is $W = 100$: $\frac{X}{1 - X} = 3$ fixes $\frac{kC_{A0}^2}{F_{A0}} = \frac{3}{100} = 0.03 \text{ kg}^{-1}$.

Step 4 — with drop. $W - \frac{\alpha W^2}{2} = 100 - 49.5 = 50.5$:

$$\frac{X}{1 - X} = 0.03 \times 50.5 = 1.515 \Rightarrow X = 0.60.$$

Pressure drop has cost $0.75 - 0.60 = 0.15$ of conversion.

The slide does not print C_{A0} and F_{A0} separately; the combined group 0.03 kg^{-1} is implied by its own $\alpha = 0$ answer and is stated here instead of being invented.

$X = 0.60$ with pressure drop, 0.75 without.

QQ19 – PBR + CSTR in series. Ch4, slides 76–77

Source: Ch4_Krd.pdf, slides 76–77.

First order in p_A , $A \rightarrow B$ (gas). A PBR with 50 kg of catalyst gives 50% conversion at $P_0 = 20 \text{ atm}$ with $\alpha = 0.018 \text{ kg}^{-1}$. A CSTR holding the same 50 kg is added in series. (a) Which order gives the higher conversion? (b) The exit conversion; (c) the pressure at the PBR exit; (d) the effect of halving the catalyst diameter and enlarging the pipe 50% (turbulent flow).

Set-up — the coupled ODEs. First order in p_A with expansion: $\frac{dX}{dW} = a \frac{1 - X}{1 + \varepsilon X} p$ together with the Ergun (simplified) drop $\frac{dp}{dW} = -\frac{\alpha}{2p}$; shooting them to match $X(50) = 0.5$ fixes $a = kP_0/F_{A0} = 0.0247 \text{ kg}^{-1}$.

(c) The PBR exit pressure. $p = \sqrt{1 - \alpha W} = \sqrt{1 - 0.9} = 0.316$, so $P = 20 \times 0.316 = 6.3 \text{ atm}$ — the bed eats two-thirds of the pressure.

(a)+(b) Which order first. Integrate the pair through the PBR then solve the CSTR balance (and the reverse order): PBR→CSTR gives $X = 0.60$; CSTR→PBR gives $X = 0.69$. Putting the *CSTR first* is better: it runs on undropped pressure and leaves the pressure-sensitive PBR the full 20 atm.

(d) Smaller pellets, fatter pipe, turbulent flow. Then $\alpha \propto D_p^{-1} D_{\text{pipe}}^{-4}$: $\alpha' = 0.018 \times 2 \times 1.5^{-4} = 0.0071 \text{ kg}^{-1}$; less drop means the PBR works harder, and PBR→CSTR rises to $X \approx 0.73$ (the other order rises similarly).

Exam one-liner for (a): “a CSTR cares only about the pressure at its own inlet; a PFR/PBR integrates it all along — so spend the pressure drop last.”

(a) CSTR upstream; (b) $X \approx 0.69$ (0.60 the other way); (c) 6.3 atm; (d) $X \approx 0.73$.

QQ20 – Series reactions in a batch reactor. Ch4, slides 78–79

Source: Ch4_Krd.pdf, slides 78–79.

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$ with $k_1 = 1 \text{ h}^{-1}$, $k_2 = 2 \text{ h}^{-1}$, $C_A(0) = 5 \text{ mol/m}^3$, $C_B(0) = C_C(0) = 0$. Give the profiles and the maximum of B.

Step 1 — A decays alone. $C_A = 5e^{-k_1 t} = 5e^{-t}$.

Step 2 — B is made and consumed. $\frac{dC_B}{dt} = k_1 C_A - k_2 C_B$; the standard integrating-factor result:

$$C_B = 5 \frac{k_1}{k_2 - k_1} \left(e^{-k_1 t} - e^{-k_2 t} \right) = 5 \left(e^{-t} - e^{-2t} \right).$$

Step 3 — C by closure. $C_C = 5 - C_A - C_B$.

Step 4 — the maximum of the intermediate. $dC_B/dt = 0$ gives

$$t_{max} = \frac{\ln(k_2/k_1)}{k_2 - k_1} = \ln 2 = 0.693 \text{ h}, \quad C_B^{max} = 5(0.5 - 0.25) = 1.25 \text{ mol/m}^3.$$

The shape to sketch in the exam: A falls exponentially; B rises through its peak at t_{max} then falls; C is the monotone mirror image. Series intermediates *always* pass through a maximum.

$$t_{max} = \ln 2 \approx 0.69 \text{ h}, \quad C_B^{max} = 1.25 \text{ mol/m}^3; \text{ profiles as above.}$$

QQ21 – PFR pressure drop: pipe length. Ch4, slides 90–98

Source: Ch4_Krd.pdf, slides 90–98.

$A \rightarrow B$, first order, $P_0 = 10 \text{ atm}$, $Q_0 = 0.005 \text{ m}^3/\text{s}$, 300 K , $k = 0.1 \text{ s}^{-1}$, $M = 60 \text{ g/mol}$, $\mu = 10^{-5} \text{ Pa} \cdot \text{s}$. Pipe length for (a) 10% in a 2-cm pipe; (b) 10% in a 1.5-cm pipe; (c) 20% in the 1.5-cm pipe?

Step 1 — enter the flow. $\rho_0 = P_0 M / RT = 24.4 \text{ kg/m}^3$; in the 2-cm pipe $v_0 = Q_0 / A = 0.005 / 3.14 \times 10^{-4} = 15.9 \text{ m/s}$; $F_{A0} = 2.03 \text{ mol/s}$.

Step 2 — the no-drop benchmark. Without drop, $L_{10} = \frac{v_0}{k} \ln \frac{1}{0.9} = 16.8 \text{ m}$ and $L_{20} = 35.5 \text{ m}$ in the 2-cm pipe — pressure drop can only make these longer.

Step 3 — with drop, the deck's coupled ODEs. The deck integrates dX/dz and dP/dz together (slides 92–95): (a) about 20 m for 10% in the 2-cm pipe.

(b) 1.5-cm pipe, 10%. The smaller bore raises v_0 (to about 28 m/s) and steepens the pressure fall, but 10% is still reachable — at a longer length.

(c) 1.5-cm pipe, 20%. Not possible: the pressure collapses towards zero before 20% is reached (slide 98: “10% conversion is possible, but 20% is not”) — once $P \rightarrow 0$ the velocity diverges and the reaction starves.

(a) $\approx 20 \text{ m}$; (b) 10% achievable; (c) 20% unattainable in the 1.5 cm pipe.

Part E – Collection and analysis of rate data (Ch5_krd.pdf)

QQ22 – Trityl + methanol: order and rate constant. Ch5, slides 12–17

Source: Ch5_

,krd.pdf, slides 12–17 (and 42–49).

$(\text{C}_6\text{H}_5)_3\text{CCl} + \text{CH}_3\text{OH} \rightarrow (\text{C}_6\text{H}_5)_3\text{COCH}_3 + \text{HCl}$ in a batch reactor; $C_{B0} = 0.5 \text{ mol/dm}^3$ (methanol) against trityl ($\times 10^3 \text{ mol/dm}^3$) = 50, 38, 30.6, 25.6, 22.2, 19.5, 17.4 at $t = 0, 50, \dots, 300 \text{ min}$. (1) Find the order in trityl; (2) the rate constant, given first order in methanol.

Set-up — pseudo-order. Methanol is in ten-fold excess, so $C_B \approx C_{B0}$ throughout and $-r_A = kC_A^\alpha C_B \approx (kC_{B0})C_A^\alpha = k'C_A^\alpha$: the runs see a single unknown α .

Step 1 — differential method (slide 17). Estimate $-dC_A/dt$ at each point by finite differences (deck values, $\times 10^{-4} \text{ mol}/(\text{dm}^3 \cdot \text{min})$): 2.86, 1.94, 1.24, 0.84, 0.61, 0.48, 0.36. Plot $\ln(-r_A)$ vs $\ln C_A$: slope = $1.99 \approx 2$ — *second order in trityl*.

Step 2 — integral check. If $\alpha = 2$, $\frac{1}{C_A} - \frac{1}{C_{A0}} = k't$ must be a straight line through the origin; it is, with slope $k' = 0.124 \approx 0.125 \text{ dm}^3/(\text{mol} \cdot \text{min})$. Two independent methods agreeing is the deck's point.

Step 3 — split off the methanol order. With $-r_A = kC_A^2 C_B$ and $C_B = 0.5$: $k = k'/C_{B0} = 0.125/0.5 = 0.25 \text{ dm}^6/(\text{mol}^2 \cdot \text{min})$. (The deck's nonlinear-regression slides return $\alpha \approx 2$ with the same k' .)

Exam wording: “differential method gives the order from the log–log slope; the integral method confirms it and supplies k' ; the excess reactant converts k' back to k .”

Second order in trityl; $k' = 0.125 \text{ dm}^3/(\text{mol} \cdot \text{min})$, $k = 0.25 \text{ dm}^6/(\text{mol}^2 \cdot \text{min})$.

QQ23 – Dimethyl ether decomposition bomb. Ch5, slide 75

Source: Ch5_

,krd.pdf, slide 75 (“F 5.8”).

$(\text{CH}_3)_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2 + \text{CO}$ at 504°C in a constant-volume bomb; total pressure π (mmHg) = 408, 488, 562, 799

at $t = 390, 777, 1195, 3155$ s and 931 at $t = \infty$. (a) Why is π_0 not recorded, and what was it? (b) Find the order and the rate constant.

(a) The missing first reading. The bomb had to be heated to 504°C before readings could start, and it was already reacting during that time — so π_0 was never observed. Recover it from stoichiometry: 1 mol of ether gives 3 mol of product, so at complete conversion $\pi_\infty = 3\pi_0$, hence

$$\pi_0 = \frac{931}{3} \approx 310 \text{ mmHg}.$$

(b) Express p_A in terms of π . If x (pressure units) of ether has reacted, $\pi = \pi_0 + 2x$ and $p_A = \pi_0 - x$; eliminating x (and using $3\pi_0 = \pi_\infty$):

$$p_A = \frac{3\pi_0 - \pi}{2} = \frac{\pi_\infty - \pi}{2} = 261.5, 221.5, 184.5, 66.0 \text{ mmHg}.$$

Test first order: $\ln(p_{A0}/p_A)/t$ gives 4.4, 4.3, 4.3, $4.9 \times 10^{-4} \text{ s}^{-1}$ — constant to reading error, so the decomposition is first order with $k \approx 4.4 \times 10^{-4} \text{ s}^{-1}$.

The same trick — “ p_A from the total pressure using $p_{A,\infty}$ ” — is used in Q27, Q28 and Q35; learn it once.

$\pi_0 \approx 310 \text{ mmHg}$; first order, $k \approx 4.4 \times 10^{-4} \text{ s}^{-1}$.

QQ24 – First-order reversible $A \rightleftharpoons R$. Ch5, slide 72

Source: Ch5_

,krd.pdf, slide 72.

$C_{A0} = 0.5 \text{ mol/L}$, $C_{R0} = 0$. After 8 min the conversion is 33.3%; the equilibrium conversion is 66.7%. Find the rate equation.

The integrated law to use. For $-r_A = k_1 C_A - k_2 C_R$ (both directions first order) the batch integral is

$$\ln \frac{X_e}{X_e - X} = (k_1 + k_2) t,$$

so one kinetic run plus the equilibrium conversion fixes $k_1 + k_2$.

Step 1 — the sum. $\ln \frac{0.667}{0.667 - 0.333} = \ln 2 = 0.693 = (k_1 + k_2)(8)$, so $k_1 + k_2 = 0.0866 \text{ min}^{-1}$.

Step 2 — the ratio. At equilibrium the net rate vanishes: $K_e = \frac{C_{Re}}{C_{Ae}} = \frac{X_e}{1 - X_e} = \frac{0.667}{0.333} = 2 = \frac{k_1}{k_2}$.

Step 3 — solve. $3k_2 = 0.0866$: $k_2 = 0.0289$, $k_1 = 0.0577 \text{ min}^{-1}$, hence

$$-r_A = 0.058 C_A - 0.029 C_R \quad \text{mol}/(\text{L} \cdot \text{min}).$$

Check before writing the answer: $k_1/k_2 = 2 = K_e$ and $k_1 + k_2 = 0.087$ — both constraints must hold.

$-r_A = 0.058$
 $, C_A - 0.029$
 $, C_R \text{ mol}/(\text{L} \cdot \text{min}).$

QQ25 – Half-order batch; $2A \rightarrow B$ volume change. Ch5, slide 73

Source: Ch5_

,krd.pdf, slide 73.

(i) $A \rightarrow R$ with $-r_A = 3C_A^{0.5} \text{ mol}/(\text{L} \cdot \text{h})$, $C_{A0} = 1 \text{ mol/L}$: what is the conversion after 1 h? (ii) Gaseous $2A \rightarrow B$ (feed 80% A, 20% inerts) at constant pressure: the volume falls 20% in 3 min; find the first-order k .

(i) Integrate the half-order law. $\frac{dC_A}{dt} = -3C_A^{1/2} \Rightarrow 2(\sqrt{C_{A0}} - \sqrt{C_A}) = 3t$, i.e. $\sqrt{C_A} = 1 - 1.5t$. Blindly inserting $t = 1$ gives $\sqrt{C_A} = -0.5$ — impossible. The charge is *fully consumed* when $\sqrt{C_A} = 0$, at $t = \frac{2}{3} \text{ h} = 40 \text{ min}$; after that the rate is zero. Hence at 1 h, $X = 1$.

(ii) Read X off the volume change. $y_{A0} = 0.8$, $\varepsilon_A = y_{A0} \frac{1-2}{2} = -0.4$; constant T, π gives $V/V_0 = 1 + \varepsilon_A X$:

$$0.8 = 1 - 0.4X \Rightarrow X = 0.5.$$

First order: $\ln \frac{1}{1-X} = kt \Rightarrow k = \frac{\ln 2}{3} = 0.231 \text{ min}^{-1}$.

Part (i) is a trap question: half-order kinetics run out of reactant in finite time; state the depletion time explicitly in the exam.

(i) $X = 1$ (complete at 40 min); (ii) $k = 0.231 \text{ min}^{-1}$.

QQ26 – $A \rightarrow 3R$ in a MFR (L 5.19). Ch5, slide 74

Source: Ch5_-
krd.pdf, slide 74.

Pure A at 3 atm and 30 C ($C_{A0} = 120$ mmol/L) enters a 1-L mixed flow reactor; four runs with $v_0 = 0.06, 0.48, 1.5, 8.1$ L/min give $C_A = 30, 60, 80, 105$ mmol/L. Find the rate equation.

Step 1 — conversions with expansion. $A \rightarrow 3R$: $\varepsilon_A = 2$, so $C_A = C_{A0} \frac{1-X}{1+2X}$, inverted: $X = \frac{C_{A0}-C_A}{C_{A0}+2C_A} = 0.50, 0.25, 0.143, 0.045$. (Ignoring ε here would halve the first conversion — a classic slip.)

Step 2 — one rate per run from the MFR balance. $-r_A = \frac{v_0 C_{A0} X}{V} = 3.6, 14.4, 25.7, 44.0$ mmol/(L·min).

Step 3 — order and constant. The log-log slope of the four $(C_A, -r_A)$ pairs is 2.00; dividing, $-r_A/C_A^2 = 0.0040$ in all runs, so

$$-r_A = 0.004 C_A^2 \quad (\text{mmol, L, min}) = 4 \text{ L}/(\text{mol} \cdot \text{min}) \text{ in molar units.}$$

$-r_A = 0.004$
 C_A^2 with C_A in mmol/L (i.e. $k = 4 \text{ L}/(\text{mol} \cdot \text{min})$).

QQ27 – $A \rightarrow 2.5R$ batch, partial-pressure data (L 3.25). Ch5, slide 76

Source: Ch5_-
krd.pdf, slide 76.

At 0 C, constant volume, pure A: p_A (mm) = 760, 600, 475, 380, 320, 275, 240, 215 at $t = 0, 2, \dots, 14$ min and 150 at $t = \infty$. Find a rate equation that represents the data.

Step 1 — the tail is not zero: equilibrium. $p_{A,\infty} = 150 \neq 0$: the reaction stops short, so it is reversible and the rate must be tested as a first-order *approach to equilibrium*:

$$\ln \frac{p_{A0} - p_{Ae}}{p_A - p_{Ae}} = kt.$$

Step 2 — evaluate at each reading. With $p_{A0} - p_{Ae} = 610$ the eight values of k come out 0.152–0.162 min^{-1} (least-squares fit 0.160) — flat enough to accept the form.

Conclusion. A first-order reversible rate equation (forward and reverse each first order) represents the data with $k_1 + k_2 \approx 0.16 \text{ min}^{-1}$.

Same family as Q24: whenever a batch run levels off above zero, test $\ln[(p_0 - p_e)/(p - p_e)]$ vs t before anything else.

First-order approach to $p_{Ae} = 150$ mm with $k \approx 0.16 \text{ min}^{-1}$.

QQ28 – Reaction bomb $2A \rightarrow B$ at 100 C. Ch5, slides 78–79

Source: Ch5_-
krd.pdf, slides 78–79.

Pure A at 1 atm and 25 C is jumped to 100 C; the total pressure falls from 1.14 atm at $t = 1$ min to 0.728 atm at $t = 20$ min (readings e.g. 1.04, 0.982, 0.940, 0.905, 0.870 at $t = 2.6$); after the weekend no A remains. Find the rate equation in mol, L, min.

Step 1 — stoichiometry to $p_A(\pi)$. $2A \rightarrow B$ halves the moles: $\pi = p_A + p_B$ with $p_B = \frac{1}{2}(p_{A0} - p_A)$, so $p_A = 2\pi - p_{A0}$; the deck takes $p_{A0} = 1$ atm at the reaction temperature. The “no A after the weekend” reading confirms the reaction goes to completion.

Step 2 — try the orders. Plot the twelve readings as $1/p_A$ vs t (second order) and $\ln p_A$ vs t (first order): the second-order plot is the straight one ($r^2 = 0.964$ against 0.886), so the reaction is second order in A with $k_p = 0.074 \text{ atm}^{-1} \text{ min}^{-1}$ from the $1/p_A$ slope.

Step 3 — convert to concentration units. With $p_A = C_A RT$ and $RT = 0.082 \times 373 = 30.6 \text{ L} \cdot \text{atm}/\text{mol}$ at 100 C, a second-order k scales as

$$k_C = k_p RT = 0.074 \times 30.6 \approx 2.3 \text{ L}/(\text{mol} \cdot \text{min}).$$

Exam sentence: “the order is decided by which linearisation is linear; the units conversion $k_C = k_p(RT)^{1-n}$ then moves between pressure and concentration forms” ($1 - n = -1$ here).

Second order in A; $k \approx 0.074 \text{ atm}^{-1} \text{ min}^{-1} \approx 2.3 \text{ L}/(\text{mol} \cdot \text{min})$ at 100 C.

QQ29 – The gambling tables. Ch5, slide 80

Source: Ch5_-
krd.pdf, slide 80.

Raju plays at two tables at once; losses at each table are proportional to the money he holds — table 1 halves it

in 4 h, table 2 in 2 h. Starting with Rs. 1000 at both, when is Rs. 10 left?

Translate to kinetics. “Loss proportional to money held” is first-order decay: $k_1 = \ln 2/4$ and $k_2 = \ln 2/2 \text{ h}^{-1}$. Playing at both tables simultaneously is two parallel first-order reactions, so the money obeys

$$\frac{dM}{dt} = -(k_1 + k_2)M, \quad k = k_1 + k_2 = \frac{\ln 2}{4} + \frac{\ln 2}{2} = 0.520 \text{ h}^{-1}.$$

Integrate.

$$t = \frac{1}{k} \ln \frac{M_0}{M} = \frac{\ln(1000/10)}{0.520} = \frac{4.605}{0.520} = 8.9 \text{ h}.$$

The teaching point of the slide: parallel first-order paths add their rate constants — exactly the series/parallel rule used later for reactions.

About 8.9 hours.

QQ30 – Second order: bigger MFR, equal PFR. Ch5, slide 81

Source: Ch5-
,krd.pdf, slide 81.

$-r_A = kC_A^2$ gives 50% conversion in a mixed flow reactor. (i) What conversion if the reactor is replaced by one of 6 times the volume? (ii) What conversion if it is replaced by a PFR of the same size?

Step 0 — calibrate with the original reactor. MFR at $X = 0.5$:

$$kC_{A0}\tau = \frac{X}{(1-X)^2} = \frac{0.5}{0.25} = 2.$$

This single dimensionless group is all the data the question gives.

(i) **Six-fold volume.** $kC_{A0}\tau' = 12$: $\frac{X}{(1-X)^2} = 12 \Rightarrow X = 0.75$ (solve the quadratic $12(1-X)^2 = X$; the root below 1).

(ii) **Same-size PFR.** The PFR design equation for second order gives $kC_{A0}\tau = \frac{X}{1-X} = 2 \Rightarrow X = \frac{2}{3} \approx 0.67$.

Notice the contrast with Q2: six-fold the MFR volume buys only $0.5 \rightarrow 0.75$, while switching reactor *type* at equal volume buys $0.5 \rightarrow 0.67$ — second-order kinetics punish the MFR’s exit-rate operation.

(i) $X = 0.75$; (ii) $X = 0.67$.

QQ31 – $A + 3B \rightarrow 6R$ with inerts. Ch5, slide 82

Source: Ch5-
,krd.pdf, slide 82.

Consider a feed $C_{A0} = 100$, $C_{B0} = 200$, $C_{I0} = 100$ to a steady-flow reactor. The isothermal gas-phase reaction is $A + 3B \rightarrow 6R$. If $C_A = 40$ at the reactor exit, what are C_B , X_A and X_B there?

Step 1 — the expansion factor. Gas phase, so moles matter: $y_{A0} = \frac{100}{400} = 0.25$, $\delta = 6 - 4 = 2$, hence $\varepsilon_A = y_{A0}\delta = 0.5$ — the stream expands as it reacts.

Step 2 — X_A from the exit concentration. $C_A = C_{A0} \frac{1-X}{1+\varepsilon X}$ inverted:

$$X_A = \frac{C_{A0} - C_A}{C_{A0} + \varepsilon C_A} = \frac{100 - 40}{100 + 0.5(40)} = \frac{60}{120} = 0.5.$$

(A constant-density reading, $X = 0.6$, ignores the expansion and is wrong for a gas.)

Step 3 — X_B . B reacts three times faster per mole fed:

$$X_B = \frac{b}{a} \frac{C_{A0}X_A}{C_{B0}} = \frac{3 \times 100 \times 0.5}{200} = 0.75.$$

Step 4 — C_B at the expanded volume.

$$C_B = C_{B0} \frac{1 - X_B}{1 + \varepsilon X_A} = \frac{200 \times 0.25}{1.25} = 40.$$

An earlier revision of these notes dropped ε and printed $0.6/0.9/20$; the slide says *gas-phase*, and Levenspiel’s own method keeps the $(1 + \varepsilon X)$ factor — hence $0.5/0.75/40$.

$X_A = 0.5$, $X_B = 0.75$, $C_B = 40$.

QQ32 – Gaseous feed with T and π change. Ch5, slide 83

Source: Ch5.-
,krd.pdf, slide 83.

Given a gaseous feed $T_0 = 1000$ K, $\pi_0 = 5$ atm, $C_{A0} = 100$, $C_{B0} = 200$; $A + B \rightarrow 5R$; at the exit $T = 400$ K, $\pi = 4$ atm, $C_A = 20$. Find X_A , X_B , C_B .

Step 1 — ε and the temperature/pressure factor. $y_{A0} = \frac{1}{3}$, $\delta = 5 - 2 = 3$, $\varepsilon_A = 1$. Cooling *concentrates* the gas and the pressure drop dilutes it:

$$\frac{T_0}{T} \cdot \frac{\pi}{\pi_0} = \frac{1000}{400} \times \frac{4}{5} = 2.5 \times 0.8 = 2.$$

Step 2 — X_A . $C_A = C_{A0} \frac{1-X}{1+X} \times 2 = 200 \frac{1-X}{1+X} = 20$, so $\frac{1-X}{1+X} = 0.1 \Rightarrow 1 - X = 0.1 + 0.1X \Rightarrow X_A = \frac{0.9}{1.1} = 0.82$.

Step 3 — X_B . $X_B = \frac{C_{A0}X_A}{C_{B0}} = \frac{100 \times 0.818}{200} = 0.41$.

Step 4 — C_B . Same factor as A:

$$C_B = C_{B0} \frac{1 - X_B}{1 + X_A} \times 2 = \frac{200 \times 0.591}{1.818} \times 2 \approx 130.$$

Sanity check to quote in the exam: at $X = 0$ the exit concentration would be $100 \times 2 = 200$ (cold, compressed gas); reaching 20 therefore needs a large conversion — 0.82, not a small one.

$$X_A \approx 0.82, X_B \approx 0.41, C_B \approx 130.$$

QQ33 – Sucrase kinetics, Michaelis–Menten. Ch5, slide 84

Source: Ch5.-
,krd.pdf, slide 84.

Sucrose (A) hydrolysed by sucrase (S): $C_{A0} = 1.0$, $C_{S0} = 0.01$ mmol/L; batch data C_A (mmol/L) = 0.68, 0.60, 0.55, 0.38, then 0.27, 0.16, 0.09, 0.04, then 0.018, 0.006, 0.0025 at $t = 1..11$ h. Can the data be fitted by Michaelis–Menten kinetics? Evaluate k_3 and C_M .

The model and its batch integral. $-r_A = \frac{k_3 C_{S0} C_A}{C_A + C_M}$; separating variables and integrating (the deck’s “integral method”):

$$k_3 C_{S0} t = (C_{A0} - C_A) + C_M \ln \frac{C_{A0}}{C_A}.$$

Linearise. Dividing by $\ln(C_{A0}/C_A)$ gives a straight-line form, $y = mx + b$ with $x = \frac{C_{A0} - C_A}{\ln(C_{A0}/C_A)}$, $y = \frac{t}{\ln(C_{A0}/C_A)}$:

$$\frac{t}{\ln(C_{A0}/C_A)} = \frac{1}{k_3 C_{S0}} \cdot \frac{C_{A0} - C_A}{\ln(C_{A0}/C_A)} + \frac{C_M}{k_3 C_{S0}}.$$

Fit the eleven points. Least squares gives slope = 3.12 h and intercept = 1.61 h with $r^2 > 0.99$ — the points lie on the line, so the Michaelis–Menten form *does* represent the data. Hence $k_3 C_{S0} = 1/3.12 = 0.321$ h⁻¹, i.e. $k_3 = 0.321/0.01 \approx 32$ h⁻¹, and $C_M = 1.61 \times 0.321 \approx 0.52$ mmol/L.

Why this linearisation: both parameters appear linearly, so one plot tests the model *and* extracts k_3, C_M — the standard exam move for Michaelis–Menten batch data.

$$\text{Yes: } k_3 \approx 32 \text{ h}^{-1} \text{ (at } C_{S0} = 0.01), C_M \approx 0.5 \text{ mmol/L.}$$

QQ34 – Reversible PFR: equilibrium then actual conversion. Ch5, slide 61

Source: Ch5.-
,krd.pdf, slide 61.

A PFR of 2 m³ processes an aqueous feed (100 L/min) with $C_{A0} = 100$ mmol/L; $A \rightleftharpoons B$. Find the equilibrium conversion, then the actual conversion — and repeat for a CSTR.

The rate law the slide implies. The slide states the problem without constants; the textbook original it reproduces (Levenspiel, whose companion MFR version is printed with $-r_A = 0.04C_A - 0.01C_R$ mol/(L·min)) supplies them; the deck's own answers below confirm the pairing.

Step 1 — equilibrium conversion. $K_e = k_1/k_2 = 0.04/0.01 = 4$, and with $C_R = C_{A0} - C_A$: $K_e = \frac{X_e}{1-X_e} \Rightarrow X_e = \frac{4}{5} = 0.80$.

Step 2 — one clean rate form. Substituting $C_R = C_{A0} - C_A$: $-r_A = (k_1 + k_2)C_A - k_2C_{A0} = 0.05(C_A - 0.02)$ — first order towards the equilibrium concentration $C_{Ae} = 0.02$.

Step 3 — PFR, $\tau = V/v_0 = 2000/100 = 20$ min.

$$\tau = \int_{C_{Af}}^{C_{A0}} \frac{dC_A}{0.05(C_A - 0.02)} \Rightarrow \ln \frac{0.08}{C_{Af} - 0.02} = 0.05 \times 20 = 1,$$

so $C_{Af} - 0.02 = 0.08/e = 0.0294$, $C_{Af} = 0.0494$, $X_{PFR} = 0.51$.

Step 4 — CSTR for comparison. $0.05(C_A - 0.02) \times 20 = C_{A0} - C_A \Rightarrow C_A - 0.02 = 0.1 - C_A \Rightarrow C_A = 0.06$, $X_{CSTR} = 0.40$ — the PFR beats the CSTR towards equilibrium, as always for a simple reversible reaction.

$X_e = 0.80$; PFR $X \approx 0.51$; CSTR $X \approx 0.40$ (rate law quoted from the textbook original, as the slide omits it).

QQ35 – $A \rightarrow R + 2S$ from total-pressure data. Ch5, slide 62

Source: Ch5_

,krd.pdf, slide 62.

Pure A at constant volume; total pressure π (mmHg) = 7.5, 10.5, 12.5, 15.8, 17.9, 19.4 at $t = 0, 2.5, 5, 10, 15, 20$ min. Find the rate equation.

Step 1 — $p_A(\pi)$. Per mole A reacted the moles rise by $\Delta n = (1 + 2) - 1 = 2$: $\pi = \pi_0 + 2x$, $p_A = \pi_0 - x$, hence

$$p_A = \pi_0 - \frac{\pi - \pi_0}{2} = \frac{3\pi_0 - \pi}{2} = 7.5, 6.0, 5.0, 3.35, 2.3, 1.55.$$

Step 2 — test first order. $\ln(p_{A0}/p_A)/t$ gives 0.089, 0.081, 0.081, 0.079, 0.079 min⁻¹ — constant within scatter: first order with $k \approx 0.08$ min⁻¹.

Here π_0 is recorded (7.5), unlike Q23; the same $p_A = (3\pi_0 - \pi)/2$ algebra covers every 1 → 3-type expansion in this part.

First order with $k \approx 0.08$ min⁻¹.

Part F – Multiple reactions (Ch6_krd.pdf)

QQ36 – Product distribution, three contacting patterns. Ch6, slide 32

Source: Ch6_krd.pdf, slide 32.

Aqueous $A + B \xrightarrow{k_1} R$ (desired) and $A + B \xrightarrow{k_2} S$ with $r_R = 1.0 C_A^{1.5} C_B^{0.3}$, $r_S = 1.0 C_A^{0.5} C_B^{1.8}$ mol/(L·min). Equal streams of A and B at 20 mol/L each are fed (after mixing $C_{A0} = C_{B0} = 10$); 90% conversion of A is required. Find C_R for (a) plug flow, (b) mixed flow, (c) plug flow with side entrances of B.

Step 1 — the instantaneous fraction. With $C_A = C_B = c$ throughout (equal feeds, equal consumption),

$$\varphi(R/A) = \frac{r_R}{r_R + r_S} = \frac{c^{1.5} c^{0.3}}{c^{1.5} c^{0.3} + c^{0.5} c^{1.8}} = \frac{1}{1 + c^{0.5}}.$$

φ is larger at low concentration — that single observation already ranks the three contactors.

(a) **Plug flow.** Integrate φ over c : $10 \rightarrow 1$:

$$C_R = \int_1^{10} \frac{dc}{1 + \sqrt{c}} \stackrel{u=\sqrt{c}}{=} 2[u - \ln(1 + u)]_1^{\sqrt{10}} = 2.86 \text{ mol/L}.$$

(b) **Mixed flow.** The whole reactor sits at the exit level $c = 1$: $\varphi = \frac{1}{2}$, so $C_R = \frac{1}{2}(10 - 1) = 4.50$ mol/L — better, because the CSTR keeps c low where φ is high.

(c) **Side entry of B.** Feeding B along the length holds C_B at the low exit value ($= 1$) while C_A stays high: $\varphi = \frac{c^{1.5}}{c^{1.5} + c^{0.5}} = \frac{c}{c+1}$,

$$C_R = \int_1^{10} \frac{c dc}{c+1} = 9 - \ln \frac{11}{2} = 7.30 \text{ mol/L}.$$

The deck's lesson: when the desired reaction has the lower concentration exponent, run at low concentrations (CSTR beats PFR), and side-feeding the other reactant beats both.

(a) 2.86; (b) 4.50; (c) 7.30 mol/L — mixed flow beats plug flow here, and keeping B low is best.

QQ37 – Two MFRs in series. Ch6, slide 33

Source: Ch6_krd.pdf, slide 33.

$r_R = k_1 C_A^2$, $r_S = k_2 C_A$; feed $C_{A0} = 1.0$, $C_{R0} = 0$, $C_{S0} = 0.3$; $\tau_1 = 2.5$, $\tau_2 = 10$ min; the first reactor exits at $C_A = 0.4$, $C_R = 0.2$, $C_S = 0.7$. Find the second reactor's exit.

Step 1 — extract the constants from reactor 1. MFR balances:

$$k_1 = \frac{C_{R1} - C_{R0}}{\tau_1 C_{A1}^2} = \frac{0.2/2.5}{0.4^2} = 0.5, \quad k_2 = \frac{(C_{S1} - C_{S0})/\tau_1}{C_{A1}} = \frac{0.4/2.5}{0.4} = 0.4.$$

Step 2 — A balance on reactor 2. $0.4 - C_A = (0.5C_A^2 + 0.4C_A) \times 10$, i.e. $5C_A^2 + 5C_A - 0.4 = 0$:

$$C_A = \frac{-5 + \sqrt{25 + 8}}{10} = 0.0745.$$

Step 3 — R and S balances.

$$C_R = 0.2 + 0.5C_A^2(10) = 0.228, \quad C_S = 0.7 + 0.4C_A(10) = 0.998.$$

Mass check to state in the exam: $C_A + C_R + C_S$ should equal $C_{A0} + C_{S0} = 1.3$ (R and S each come from A): $0.075 + 0.23 + 1.00 = 1.30$.)

$$C_A \approx 0.075, \quad C_R \approx 0.23, \quad C_S \approx 1.00 \text{ mol/L.}$$

Part G – Non-isothermal design (Ch7_krd.pdf)

QQ38 – Adiabatic PFR: n-butane isomerisation. Ch7, slides 14–17

Source: Ch7_krd.pdf, slides 14–17.

n-butane $\rightleftharpoons$ isobutane, adiabatic liquid-phase PFR; $k = 31.1 \text{ h}^{-1}$ at 360 K, $E = 65.7 \text{ kJ/mol}$, $\Delta H_{Rx} = -6900 \text{ J/mol}$, $K_C = 3.03$ at 60 C, $C_{A0} = 9.3 \text{ kmol/m}^3$, $C_p = 141/141/161 \text{ J/(mol}\cdot\text{K)}$ (butane/isobutane/pentane). Process 163 kmol/h (90 mol% butane) to $X = 0.70$, feed at 330 K.

Adiabatic line $T = 330 + 43.3X$ (deck); $k(T) = 31.1e^{7902(1/360 - 1/T)}$; $K_C(T) = 3.03e^{830.3(1/333 - 1/T)}$; $F_{A0} = 0.9 \times 163 = 146.7 \text{ kmol/h}$, and

$$-r_A = kC_{A0} \left[1 - \left(1 + \frac{1}{K_C} \right) X \right].$$

The deck's table (e.g. $-r_A = 39.2$ at $X = 0$, $F_{A0}/(-r_A) = 23.29 \text{ m}^3$ at 0.7) is reproduced by these expressions; integrating $F_{A0}/(-r_A)$ to $X = 0.7$ gives $V \approx 2.0 \text{ m}^3$ (the deck's coarse Simpson table reads $\approx 2.4 \text{ m}^3$; the value is sensitive near $X_e \approx 0.72$).

$$V \approx 2\text{--}2.4 \text{ m}^3 \text{ for } 70\% \text{ (deck table: } F_{A0}/(-r_A) = 23.3 \text{ m}^3 \text{ at } X = 0.7).$$

QQ39 – Adiabatic equilibrium with two interstage coolers. Ch7, slides 48–51

Source: Ch7_krd.pdf, slides 48–51.

$A \rightleftharpoons B$, pure A at 300 K; $K_C(298) = 10^5$, $\Delta H_{Rx} = -20\,000 \text{ cal/mol}$ (as implied by the deck's $K_C(T)$ and adiabatic line), $C_{pA} = C_{pB} = 50 \text{ cal/(mol}\cdot\text{K)}$, so $X_{ad} = 2.5 \times 10^{-3}(T - 300)$. Two coolers return the stream to 350 K; 95% of the equilibrium conversion in each reactor; $F_{A0} = 40 \text{ mol/s}$; cooling water at 270 K, $c_{pC} = 18 \text{ cal/(mol}\cdot\text{K)}$, $U = 100 \text{ cal/(s}\cdot\text{m}^2\cdot\text{K)}$.

$K_C(T) = 10^5 \exp[-33.78(T - 298)/T]$, $X_e = K_C/(1 + K_C)$. Reactor 1: $2.5 \times 10^{-3}(T - 300) = 0.95X_e(T) \Rightarrow T_1 = 459 \text{ K}$, $X_1 = 0.40$. Cool to 350 K. Reactor 2: $X_1 + 2.5 \times 10^{-3}(T - 350) = 0.95X_e(T) \Rightarrow T_2 = 440 \text{ K}$, $X_2 = 0.62$. Duties: $\dot{Q} = \dot{n}C_p(T_{out} - 350)$: $\dot{Q}_1 = 40 \times 50 \times 109 = 218 \text{ kcal/s}$; $\dot{Q}_2 = 40 \times 50 \times 90 = 179 \text{ kcal/s}$ (water flow = $\dot{Q}/(18\Delta T)$ as sized by the deck).

$$X \approx 0.40 \text{ after reactor 1 and } 0.62 \text{ after reactor 2; } \dot{Q}_1 \approx 218, \dot{Q}_2 \approx 179 \text{ kcal/s.}$$

QQ40 – Propylene-oxide CSTR, adiabatic. Ch7, slides 57–59

Source: Ch7_krd.pdf, slides 57–59 (and 69–72).

300-gal glass-lined CSTR; $F_{A0} = 43.04 \text{ lbmol/h}$ PO; streams: $46.62 \text{ ft}^3/\text{h}$ PO, $46.62 \text{ ft}^3/\text{h}$ methanol, $233.1 \text{ ft}^3/\text{h}$ water (0.1 wt% H_2SO_4); methanol 71.87 and water 802.8 lbmol/h ; $T_0 = 75 \text{ F} = 535 \text{ R}$ after the 17 F heat-of-mixing rise; first order in PO with $k = 16.96 \times 10^{12} e^{-32\,400/RT} \text{ h}^{-1}$; limit 125 F.

$v_0 = 326.3 \text{ ft}^3/\text{h}$, $C_{A0} = 0.132 \text{ lbmol}/\text{ft}^3$, $\tau = (300/7.48)/326.3 = 0.123 \text{ h}$. Energy balance: $\sum \Theta_i C_{p,i} \approx 404 \text{ Btu}/(\text{lbmol A} \cdot \text{R})$, $-\Delta H_{Rx} = 36\,400 \text{ Btu}/\text{lbmol} \Rightarrow T = 535 + 90.1X$. Solving with $X = \frac{k\tau}{1+k\tau}$: $X = 0.84$, $T = 611 \text{ R} = 151 \text{ F}$ (deck table intersection $\approx 613 \text{ R}$, $X \approx 0.85$). $151 > 125 \text{ F}$: the reactor *cannot* be used adiabatically.

$X \approx 0.84\text{--}0.85$ at $T \approx 611\text{--}613 \text{ R}$ ($\approx 151 \text{ F}$) $> 125 \text{ F}$ — not acceptable.

QQ41 – The same CSTR with a cooling coil. Ch7, slides 60–74

Source: Ch7-krd.pdf, slides 60–74.

A 40 ft^2 coil, $U = 100 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot \text{F})$, coolant at 85 F (545 R). Does the 125 F constraint now hold?

Energy balance adds $UA(T - T_a) = 4000(T - 545) \text{ Btu}/\text{h}$: $X = \frac{17\,379(T-535)+4000(T-545)}{43.04 \times 36\,400}$ together with the mole balance. Intersection: $X \approx 0.35\text{--}0.36$, $T \approx 562\text{--}564 \text{ R} = 102\text{--}104 \text{ F}$ — the deck's solve gives $X = 0.364$, $T = 563.7 \text{ R}$. The constraint is satisfied with the coil.

Yes: $X \approx 0.36$ at $T \approx 564 \text{ R}$ ($\approx 104 \text{ F}$) $< 125 \text{ F}$.

QQ42 – Parallel reactions in a PFR with heat effects. Ch7, slide 95

Source: Ch7-krd.pdf, slide 95.

$A \rightarrow B$ ($r_B = k_1 C_A$), $2A \rightarrow C$ ($r_C = k_2 C_A^2$); pure A at $100 \text{ mol}/\text{s}$, 150 C , $C_{A0} = 0.1 \text{ mol}/\text{dm}^3$; $\Delta H_1 = -20\,000$, $\Delta H_2 = -60\,000 \text{ J}/\text{mol A}$; $C_{pA} = C_{pB} = 90$, $C_{pC} = 180 \text{ J}/(\text{mol} \cdot \text{C})$; $E_1/R = 4000 \text{ K}$, $E_2/R = 9000 \text{ K}$; $k_1 = 10$, $k_2 = 0.09$ (as printed, per min). Determine the temperature and flow-rate profiles down the reactor.

Coupled ODEs (per the deck's PFR heat-effects balances): $\frac{dF_A}{dV} = -(r_1 + 2r_2)$, $\frac{dF_C}{dV} = r_2$, $\frac{dT}{dV} = \frac{r_1(-\Delta H_1) + r_2(-\Delta H_2)}{F_A C_{pA} + F_B C_{pB} + F_C C_{pC}}$ with $C_A = F_A/\dot{V}_0$, $\dot{V}_0 = 60\,000 \text{ dm}^3/\text{min}$ (taking the printed rate constants as per minute). Integrating to $V = 10 \text{ m}^3$: $F_A \rightarrow 0$, $F_C \rightarrow 0$ (B dominates), and T rises to $\approx 645 \text{ K}$ — the adiabatic limit for complete conversion to B ($\Delta T = 100 \times 20\,000/(100 \times 90) \approx 222 \text{ K}$ above 423 K).

Both reactions complete within the first m^3 ; $T \rightarrow \approx 645 \text{ K}$, product essentially B.

The slide states the problem and the data but no units for k_1, k_2 beyond “min”; the profile above takes them as printed (per minute). The qualitative result (fast exothermic run to the adiabatic temperature rise) is unaffected.

Part H – Catalysis (Catalysis_Krd.pdf)

QQ43 – Pellet concentration profile. Catalysis, slide 48

Source: Catalysis_Krd.pdf, slide 48 (“Problem”).

$A \rightarrow B$, first order, spherical particle $d_p = 2 \times 10^{-3} \text{ cm}$; $C_{As} = 0.001 \text{ gmol}/\text{m}^3$; $D_e = 0.1 \text{ cm}^2/\text{s}$. The concentration halfway to the centre is $1/10$ of C_{As} . (a) C_A at $3 \times 10^{-4} \text{ cm}$ from the external surface; (b) diameter for $\eta = 0.8$.

Sphere, first order: $\frac{C_A}{C_{As}} = \frac{R}{r} \frac{\sinh(\phi r/R)}{\sinh \phi}$. At $r = R/2$: $0.1 = \frac{2 \sinh(\phi/2)}{\sinh \phi} = \frac{1}{\cosh(\phi/2)} \Rightarrow \phi = 2 \operatorname{arcosh}(10) = 5.99$. (a) $r = R - 3 \times 10^{-4} = 7 \times 10^{-4} \text{ cm}$: ratio = $\frac{1}{0.7} \frac{\sinh(0.7 \times 5.99)}{\sinh 5.99} = 0.237$, $C_A = 2.4 \times 10^{-4} \text{ gmol}/\text{m}^3$. (b) $\eta = \frac{3}{\phi^2}(\phi \coth \phi - 1) = 0.8 \Rightarrow \phi_{0.8} = 2.04$; $\phi \propto d_p$: $d_{new} = 2 \times 10^{-3} \times \frac{2.04}{5.99} = 6.8 \times 10^{-4} \text{ cm}$.

(a) $C_A \approx 2.4 \times 10^{-4} \text{ gmol}/\text{m}^3$; (b) $d_p \approx 6.8 \times 10^{-4} \text{ cm}$.

QQ44 – Internal diffusion: 100 vs 150 C. Catalysis, slide 57

Source: Catalysis_Krd.pdf, slide 57.

First order, $R = 0.2 \text{ cm}$, $D_e = 0.015 \text{ cm}^2/\text{s}$; $k(100 \text{ C}) = 0.93 \text{ s}^{-1}$, $E = 20 \text{ kcal}/\text{mol}$; $C_{Ab} = 3.25 \times 10^{-5} \text{ mol}/\text{L}$. (a) rate at 100 C ; (b) at 150 C ; (c) apparent activation energy.

(a) $\phi_1 = R\sqrt{k_1/D_e} = 1.575$; $\eta_1 = 0.866$; $-r_A = \eta_1 k_1 C_{Ab} = 2.6 \times 10^{-5} \text{ mol}/(\text{L} \cdot \text{s})$. (b) $k_2 = 0.93 e^{\frac{20\,000}{1.987}(1/373 - 1/423)} = 22.5 \text{ s}^{-1}$; $\phi_2 = 7.75$; $\eta_2 = 0.337$; $-r_A = 2.5 \times 10^{-4} \text{ mol}/(\text{L} \cdot \text{s})$. (c) From the two rates, $E_{app} = R \ln(r_2/r_1)/(1/T_1 - 1/T_2) \approx 14.1 \text{ kcal}/\text{mol}$ — between E and $E/2$, moving toward the diffusion-limited $E/2 = 10 \text{ kcal}/\text{mol}$ as ϕ grows.

(a) 2.6×10^{-5} ; (b) $2.5 \times 10^{-4} \text{ mol}/(\text{L} \cdot \text{s})$; (c) $E_{app} \approx 14 \text{ kcal}/\text{mol}$.

Part I – Residence time distributions (Non_ideal_KRD.pdf)

QQ45 – Tracer pulse: $E(t)$ and the 3–6 min fraction. Non-ideal, slides 17–19

Source: Non_ideal_KRD.pdf, slides 17–19 (“Problem”).

C (g/m³) = 0, 1, 5, 8, 10, 8, 6, 4, 3, 2.2, 1.5, 0.6, 0 at $t = 0, 1, \dots, 10, 12, 14$ min. Construct $E(t)$ and find the fraction that spent 3–6 min in the reactor.

$\int_0^\infty C dt = 47.4 + 2.6 = 50.0$ g·min/m³ (Simpson on 0–10 plus 10–14, as on the deck). $E(t) = C(t)/50$: peak $E(4) = 0.20$. Fraction between 3 and 6 min: $\int_3^6 E dt \approx \frac{1}{50} \times \frac{3}{8} (8 + 3 \cdot 10 + 3 \cdot 8 + 6) = 0.465 \approx 0.5$. Mean residence time $t_m = \int tE dt \approx 5.2$ min.

$\int C dt = 50$ g·min/m³; about half the material (0.47–0.51) spent 3–6 min inside; $t_m \approx 5.2$ min.

QQ46 – Dispersion model: conversion four ways. Non-ideal, slides 52–59

Source: Non_ideal_KRD.pdf, slides 52–59 (“Problem”).

First order $A \rightarrow B$ in a 10 cm, 6.36 m tubular reactor, $k = 0.25$ min^{−1}; tracer test of Q45. Conversion by (i) closed-vessel dispersion model, (ii) PFR, (iii) single CSTR, (iv) tanks in series.

$t_m = 5.13$ min, $\sigma^2 = 5.92$ min² (deck values), $\sigma_\theta^2 = 0.225$; $\sigma_\theta^2 = \frac{2}{Pe} - \frac{2}{Pe^2}(1 - e^{-Pe}) \Rightarrow Pe = 7.7$. $Da = kt_m = 1.28$, $q = \sqrt{1 + 4Da/Pe} = 1.29$: $X = 1 - \frac{4q e^{Pe/2}}{(1+q)^2 e^{Pe q/2} - (1-q)^2 e^{-Pe q/2}} = 0.68$ (deck value). (ii) $X = 1 - e^{-Da} = 0.72$; (iii) $X = \frac{Da}{1+Da} = 0.56$; (iv) $n = t_m^2/\sigma^2 = 4.45$: $X = 1 - (1 + Da/n)^{-n} = 0.68$.

(i) 0.68; (ii) 0.72; (iii) 0.56; (iv) 0.68.

Every number above is recomputed by `krd-answers-check.py` (kept in this folder) from the data quoted here; discrepancies with a deck’s own printed answer, where any, are discussed in that question’s Notes/Warning box.